

Run 09 | NME

Global (including India)

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 69,931 (5.21%) **Case time** 56m 00s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9795). Highest F1: Random Forest (0.6655). For XGBoost, adding controls changed ROC-AUC by +0.0036 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9795	0.7139	96.21	61.93	70.86	0.6610	26s
Random Forest	0.9792	0.7166	96.27	62.44	71.24	0.6655	2m 33s
CatBoost	0.9792	0.7105	96.18	61.60	70.77	0.6587	3m 35s
H2O GBM	0.9778	0.6927	96.07	60.72	69.38	0.6476	1m 00s
Elastic Net	0.9750	0.6592	95.78	58.21	67.05	0.6232	16m 44s
Lasso	0.9750	0.6591	95.77	58.17	67.08	0.6231	14m 14s
Logistic Regression	0.9748	0.6577	95.77	58.12	66.97	0.6223	2m 02s
Ridge	0.9738	0.6412	95.36	54.05	72.19	0.6182	9m 37s
Decision Tree	0.9700	0.6053	96.13	63.35	60.85	0.6208	11s
Naive Bayes	0.9089	0.2644	89.37	29.09	72.46	0.4151	4m 34s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	17,070.52	29.18
2	Established business owner	7,587.91	12.97
3	Occupational status	3,834.35	6.55
4	Survey year	3,024.74	5.17
5	Age	2,639.81	4.51
6	Household size	1,511.82	2.58
7	Size of government	1,498.38	2.56
8	Legal system and property rights	1,487.39	2.54

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Size of government	1,498.38	2.56
2	Legal system and property rights	1,487.39	2.54
3	Regulation	1,304.18	2.23

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 09_obj3_global_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.